B.TECH CSE · MAJOR PROJECT · OCTOBER 2026

Pramana

प्रमाण *proof; a means of valid knowledge*

Citation-verified, fully local AI legal research for Indian criminal law across the IPC → BNS transition.

✓ Every citation verified | Fully offline | Zero cost

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OUTLINE

Presentation outline.

01
Introduction
who does legal research, and why it matters now

02
Problem statement
two codes, invented citations

03
Literature review
research, tools, the gap

04
Objectives
five measurable goals

05
Proposed system
facts in, checked memo out

06
Methodology
architecture, dataset, RAG, verification

07
Implementation
tech stack and live demo

08
Results
verification, benchmark, limitations

09
Conclusion
objectives met, future scope

Criminal legal research is still done by hand.

USERS

- Advocates, especially solo and district-court practitioners
- Law students
- Legal-aid clinics

1
Bare act
find candidate sections

→

2
Commentary
read the ingredients

→

3
Search judgments
keyword search for precedent

→

4
Manual reading
hours per matter
[[VERIFY: time per matter]]

WHY NOW

Two criminal codes since 1 July 2024
IPC and BNS in use side by side for years

AI tools invent citations
a wrong section is not a cosmetic error

Client facts are confidential
they should not leave the lawyer’s machine

On 1 July 2024 India renumbered its criminal law.

INDIAN PENAL CODE, 1860 [2] offences before 1 July 2024	SECTION	BHARATIYA NYAYA SANHITA, 2023 [1] offences from 1 July 2024
Punishment for murder	302	Uttering words with deliberate intent to wound religious feelings
Punishment for murder by a life-convict	303	Theft
Cheating and dishonestly inducing delivery of property	420	*No such section. The BNS ends at 358.*

A section number means nothing without its code, and *the offence date decides the code.*

AI models cite law with confidence, including law that does not exist.

A 4-BILLION-PARAMETER MODEL, ASKED ON THIS LAPTOP

Q What does BNS Section 303 cover?

A A public-panic offence: making or spreading a bomb threat.

× Wrong. BNS 303 is Theft.

20 *of* 68

citations named sections not in the BNS when the model answered from memory.

15 development cases, retrieval withheld (docs/evaluation_results.md).

17–33%

of queries hallucinated by commercial legal AI research tools [6]. [[VERIFY: Magesh et al. figure]]

PROBLEM STATEMENT

Build a research assistant for Indian criminal law that identifies applicable sections and precedents across the IPC→BNS transition, *never presents a non-existent citation*, and runs fully offline.

What exists, and what nobody combines.

RESEARCH [[VERIFY: paper details]]

[3] **RAG** · Lewis et al., NeurIPS 2020
grounds answers in retrieved text; no citation check

[4] **LeSICiN** · Paul et al., AAAI 2022
graph statute identification; supervised, IPC only

[5] **IL-TUR** · Joshi et al., ACL 2024
Indian legal benchmark; IPC-era labels

[6] **Legal AI hallucination** · Magesh et al., 2024
measures the failure, does not prevent it

[8] **BGE-M3** · Chen et al., 2024
general embedder; untested on Indian law

EXISTING TOOLS

	IPC/BNS BY DATE	VERIFIES CITATIONS	RUNS OFFLINE	FREE	
Indian Kanoon	✗	—	✗	✓	[[VERIFY]]
Manupatra / SCC Online	?	—	✗	✗	[[VERIFY]]
jhana.ai	?	✓	✗	?	[[VERIFY]]
NyayAssist	?	✓	✗	?	[[VERIFY]]
Pramana	✓	✓	✓	✓	

— search engines and databases return real documents; they do not interpret facts.

RESEARCH GAP

No system combines **offline operation**, **date-based code routing**, an **IPC→BNS precedent bridge**, **graph-verified citations** and **outcome statistics**.

Five measurable objectives.

1	Route the correct code from the offence date	→ 30 / 30 routing · slide 16
2	Identify applicable sections from the facts	→ IL-TUR benchmark · slide 17
3	Never show a non-existent section	→ 0 unverified in 889 citations · slide 16
4	Find precedents for new BNS sections	→ IPC bridge: 0 → 169 judgments · slide 13
5	Run offline on a 4 GB laptop GPU at ₹0	→ RTX 3050 Ti · slide 14

Facts and a date go in. A checked research memo comes out.

CASE FACTS · THE RUNNING EXAMPLE

The complainant parked his motorcycle outside his house at night. In the morning it was gone. CCTV shows the accused pushing it away down the lane and riding off. The accused had no permission to take it and did not return it.

OFFENCE DATE 15 Aug 2024

about a minute, offline

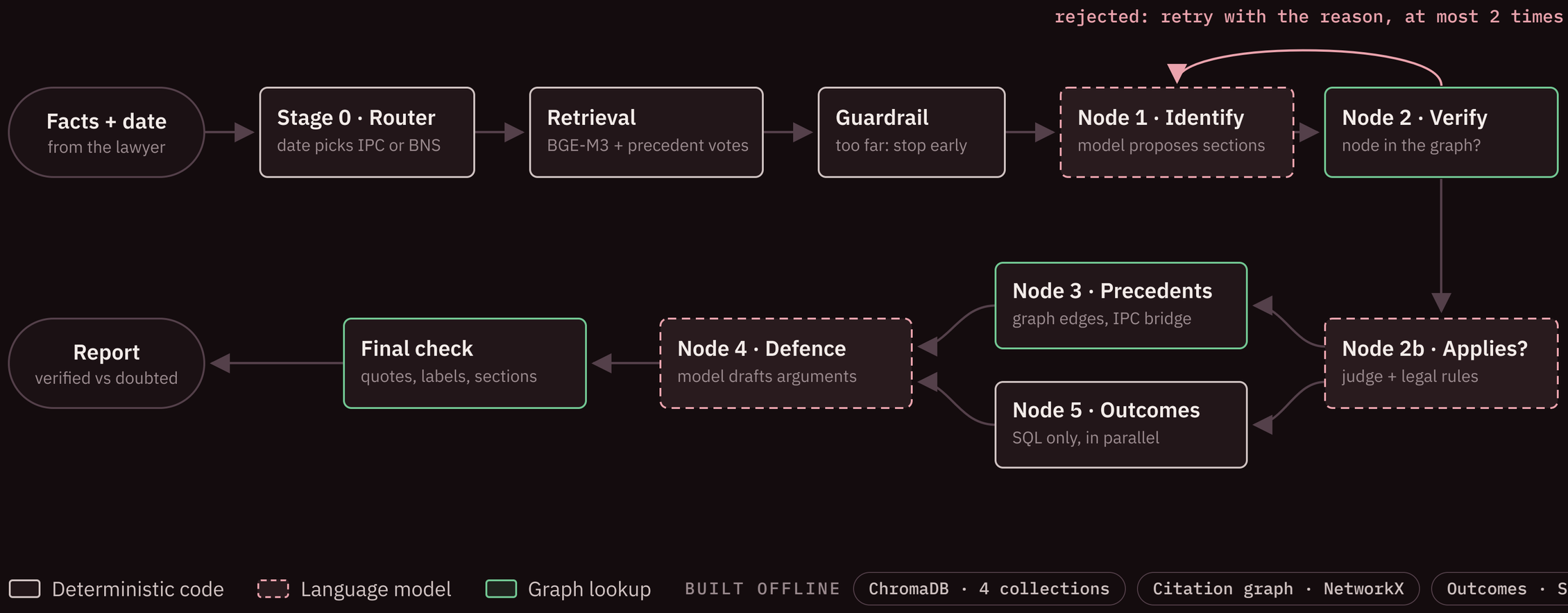
CODE	BNS, because the offence is on or after 1 July 2024
CHARGE	BNS 303 Theft ✓ verified
PRECEDENTS	3 judgments, found through its IPC predecessor, IPC 379
DEFENCE	A counter-argument, its quote checked against the facts
OUTCOMES	How 158 appeals under IPC 379 were decided

Verified citations
a section is shown only if it exists in the right code

Private by design
runs offline; client facts never leave the laptop

₹0 Free to run
open models and public data; no API key

The model proposes. The graph decides.



From public sources to three local stores.

SOURCES

IPC — 575
sections · civictech JSON + official text [2]

BNS — 358
sections · parsed from the gazette PDF [1]

Judgments — 2,699
918 Supreme Court (2023–26) + 1,781 High Court [11]

IPC→BNS map — 575
rows, each reviewed against the gazette text

PROCESSING

1 **Parse**
PDF text with PyMuPDF; repair ligatures, strip page headers

2 **Chunk**
one section per chunk; judgments by paragraph

3 **Extract**
cited sections per chunk; CrPC and other Acts filtered out

4 **Validate**
every record against a fixed schema

5 **Embed**
BGE-M3, fp16 on the GPU

STORES

ChromaDB · 4 collections
IPC sections · BNS sections · IPC-era judgments 83,918 · BNS-era 3,118
87,036 passages

Citation graph · NetworkX
933 section nodes · 2,699 judgments · 16,493 citation edges
539 IPC→BNS bridges

Outcomes · SQLite
appeal disposals per section

Retrieval finds the candidates; nothing is fine-tuned.

WHY RAG [3], NOT FINE-TUNING

Knowledge stays current
new judgments are added, not trained in

Every answer traces to a source
each candidate is a real indexed section

Fits a 4 GB GPU
no training run on the laptop

THE MOTORCYCLE CASE, STEP BY STEP

1. **Date rule**
   15 Aug 2024 ≥ 1 Jul 2024 → **BNS**

2. **Embed the facts**
   BGE-M3 [8], 1024 dimensions

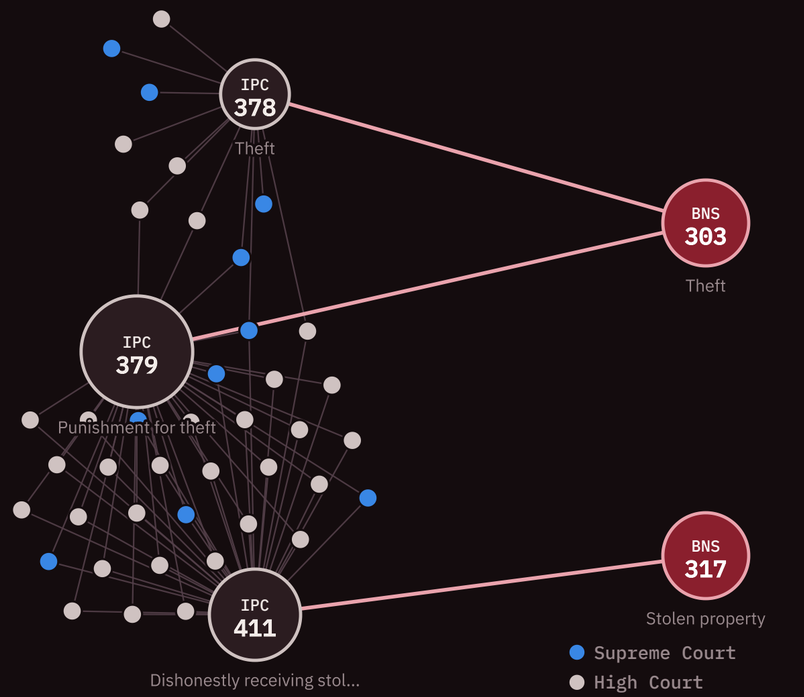
3. **Statute search + precedent votes**
   similar judgments vote for the sections they applied; fused by reciprocal rank fusion [9]

4. **14 candidates for Node 1**

303(1)	Theft	0.0320 · T3 V2
317(5)	Stolen property	0.0310 · T6 V3
324(1)	Mischief	0.0265 · T5 V30

Guardrail: closest judgment 0.686, above the 0.576 threshold, so the model is asked.

Every citation is checked against the law itself.



933 sections · 2,699 judgments · 16,493 citation edges · 539 IPC→BNS bridges

THE RETRY LOOP · A REAL RUN WITH RETRIEVAL WITHHELD

ROUND 0 — model: 421

✕ Not a section of the BNS, which has sections 1 to 358.

ROUND 1 — model: 378

✕ An IPC number. Hint from the graph: its BNS counterpart is **BNS 303**.

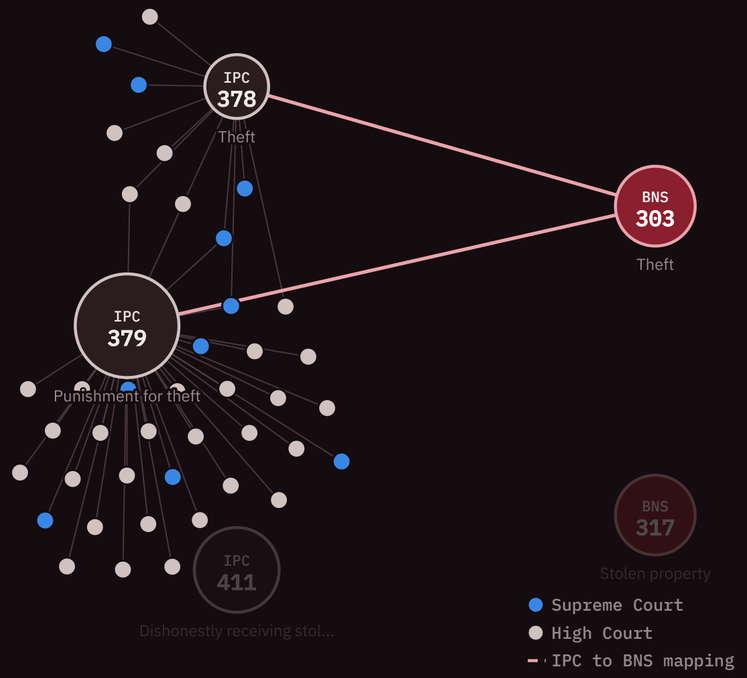
ROUND 2 — model: 303

✓ **BNS 303 · Theft.** A node of the BNS. Verified.

20 / 20
real vs invented section numbers told apart

0
rejected citations ever reached a report

New law has almost no case law yet. We bridge it to the old.



BNS 303	0	judgments cite it directly
IPC 378	11	via the mapping
IPC 379	158	via the mapping

TOP 3 BY SIMILARITY TO THE FACTS

P1	Telangana High Court	0.686	Allowed
P2	Himachal Pradesh High Court	0.676	Allowed
P3	Delhi High Court	0.658	Dismissed

Node 2b does a real section fit the facts? Doubts are shown in amber, never hidden.

Node 5 outcome counts by SQL, no LLM; fewer than 5 judgments → suppressed.

Every choice, and why.

INTERFACE

FastAPI + web app

typed endpoints, live progress streaming, serves a page that makes no outside requests

ORCHESTRATION

LangGraph [12]

vs one big prompt: a retry edge, parallel nodes, and shared state checked step by step

INTELLIGENCE

qwen3.5:4b on Ollama · **BGE-M3** [8]

vs cloud GPT / Gemini: private, ₹0, fits 4 GB

vs a legal-domain embedder: recall@3 0.533 vs 0.20, measured [10]

KNOWLEDGE

ChromaDB · **NetworkX** · **SQLite**

ChromaDB vs FAISS / Pinecone: local, persistent, metadata filters

NetworkX: exact existence check · SQLite: exact counts

ONE LAPTOP · RTX 3050 TI · 4 GB GPU



BGE-M3 encodes the facts · then the language model

Both models cannot fit at once, so every run hands the card over: unload the language model, embed, free the embedder, generate.

₹0
recurring cost

422
automated tests

Let's run it.

THE CASE

The complainant parked his motorcycle outside his house at night. In the morning it was gone. CCTV shows the accused pushing it away down the lane and riding off. The accused had no permission to take it and did not return it.

OFFENCE DATE 15 August 2024

WATCH FOR router picks BNS · 303 ranked first · verified in the graph · precedents via IPC 379

▶ LIVE

Switching to the browser

http://127.0.0.1:8000

0

invented or wrong-code sections reached a report, across every evaluation we ran.

20 / 20
citation gate: real vs invented section numbers

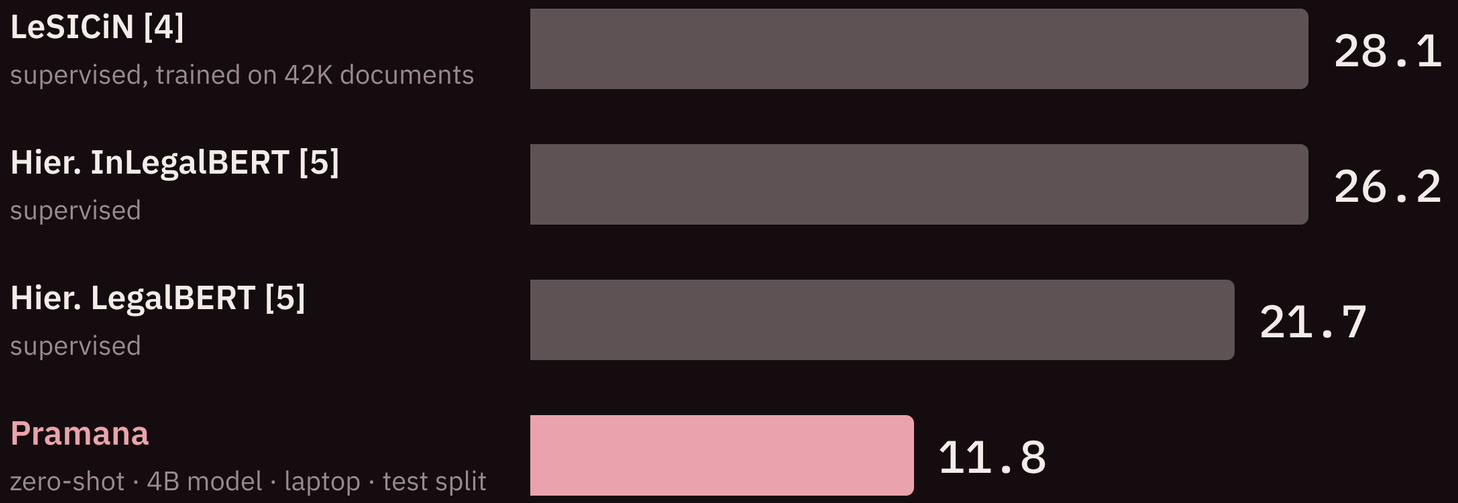
30 / 30
routing: the right code every time

889
citations checked across 263 saved runs; none unverified shown

6 / 6
cases recovered within 2 retries when the model cited from memory

Benchmarked on IL-TUR [5], with no task-specific training.

LEGAL STATUTE IDENTIFICATION · MACRO-F1 OVER 100 LABELS



100 test documents, run once after all development. Development split: 9.8. Micro precision / recall / F1: 40 / 21 / 27.

STRONG ON SERIOUS OFFENCES

302	Murder	0.78
376	Rape	0.63
34	Common intention	0.44

THE NEXT GAIN: WIDER RETRIEVAL

Minor companion offences such as simple hurt (323) or insult (504) rarely reach the candidate list: 37% of the right sections did.

What was hard, and what remains.

CHALLENGE → HOW WE HANDLED IT

No reliable digital BNS	→	parsed the official gazette PDF
Two models, one 4 GB GPU	→	hand the card over on every run
The model invents sections	→	graph verification + retry loop
No BNS case law yet	→	IPC→BNS precedent bridge

LIMITATIONS

- Zero-shot section choice trails trained systems (11.8 vs 28.1)
- Minor companion offences often missed: 37% reach the candidate list
- IPC→BNS mapping not yet reviewed by practising lawyers
- Substantive criminal law only; English text; no scanned documents

Objectives met, and where it goes next.

CONCLUSION · OBJECTIVES FROM SLIDE 7

- MET — **1 · Correct code from the date**
  30 / 30
- PARTLY — **2 · Sections from the facts**
  strong on serious offences; 11.8 macro-F1 zero-shot
- MET — **3 · No non-existent section**
  0 unverified in 889
- MET — **4 · Precedents for BNS sections**
  through the IPC bridge
- MET — **5 · Offline, 4 GB, ₹0**
  RTX 3050 Ti laptop

FUTURE SCOPE · MATCHED TO EACH LIMITATION

- ← companion offences missed
  Wider retrieval for minor offences
- ← mapping not lawyer-reviewed
  Review by practising lawyers
- ← scarce BNS case law
  Ingest new BNS judgments as published
- ← scope
  BNSS and BSA, Hindi input, scanned orders

WHO IT HELPS: Advocates · Law students · Legal-aid clinics · Firms with confidential matters

References.

[1] Government of India, “The Bharatiya Nyaya Sanhita, 2023 (Act No. 45 of 2023),” *Gazette of India*, Ministry of Law and Justice, 2023. [[VERIFY]]

[2] “The Indian Penal Code, 1860 (Act No. 45 of 1860),” Government of India. [[VERIFY]]

[3] P. Lewis *et al.*, “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in *Proc. NeurIPS*, 2020. [[VERIFY]]

[4] S. Paul, P. Goyal and S. Ghosh, “LeSICiN: A heterogeneous graph-based approach for automatic legal statute identification from Indian legal documents,” in *Proc. AAAI*, 2022. [[VERIFY]]

[5] A. Joshi *et al.*, “IL-TUR: Benchmark for Indian legal text understanding and reasoning,” in *Proc. ACL*, 2024. [[VERIFY]]

[6] V. Magesh *et al.*, “Hallucination-free? Assessing the reliability of leading AI legal research tools,” arXiv:2405.20362, 2024. [[VERIFY]]

[7] M. Dahl, V. Magesh, M. Suzgun and D. E. Ho, “Large legal fictions: Profiling legal hallucinations in large language models,” *J. Legal Analysis*, vol. 16, 2024. [[VERIFY]]

[8] J. Chen *et al.*, “BGE M3-Embedding: Multi-lingual, multi-functionality, multi-granularity text embeddings through self-knowledge distillation,” arXiv:2402.03216, 2024. [[VERIFY]]

[9] G. V. Cormack, C. L. A. Clarke and S. Büttcher, “Reciprocal rank fusion outperforms Condorcet and individual rank learning methods,” in *Proc. SIGIR*, 2009. [[VERIFY]]

[10] D. Premasiri, T. Ranasinghe and R. Mitkov, “LLM-based embedders for prior case retrieval,” in *Proc. RANLP*, 2025. [[VERIFY]]

[11] S. K. Nigam *et al.*, “NyayaAnumana and INLegalLlama: The largest Indian legal judgment prediction dataset and specialized language model,” in *Proc. COLING*, 2025. [[VERIFY]]

[12] LangChain, “LangGraph,” github.com/langchain-ai/langgraph; Ollama, ollama.com; Chroma, trychroma.com. Software, 2026. [[VERIFY]]

Thank you.

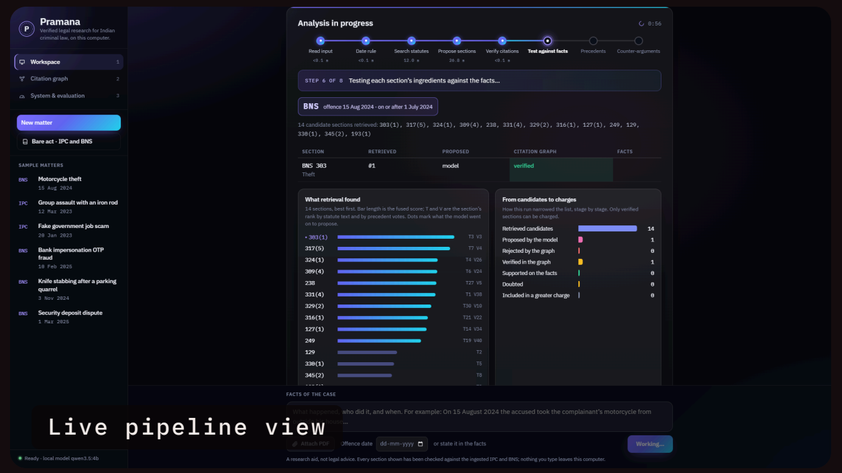
Pramana: every section it shows exists in the code the offence date selects, and it says so when it doubts one.

RAG, not fine-tuning | BGE-M3 · ChromaDB · NetworkX · LangGraph | qwen3.5:4b on Ollama | FastAPI

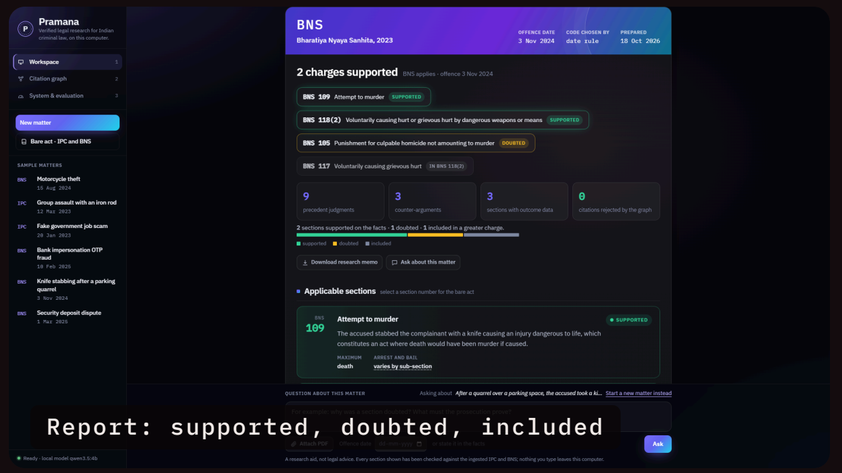
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Questions welcome.

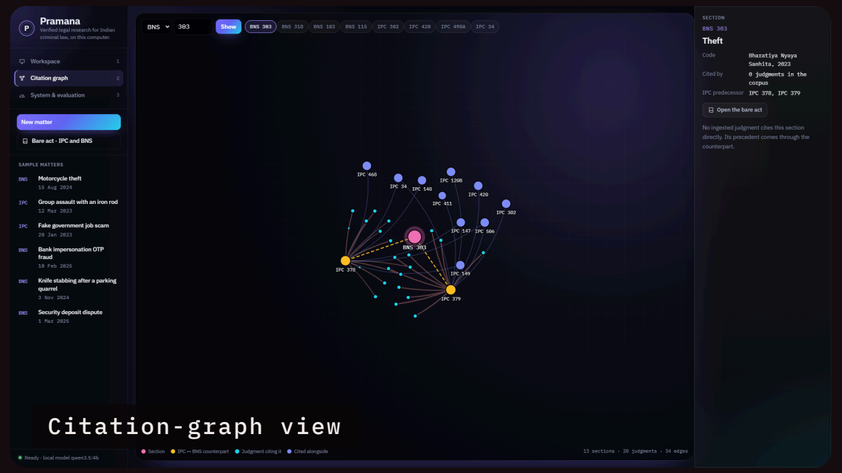
Verified and doubted are never confused.



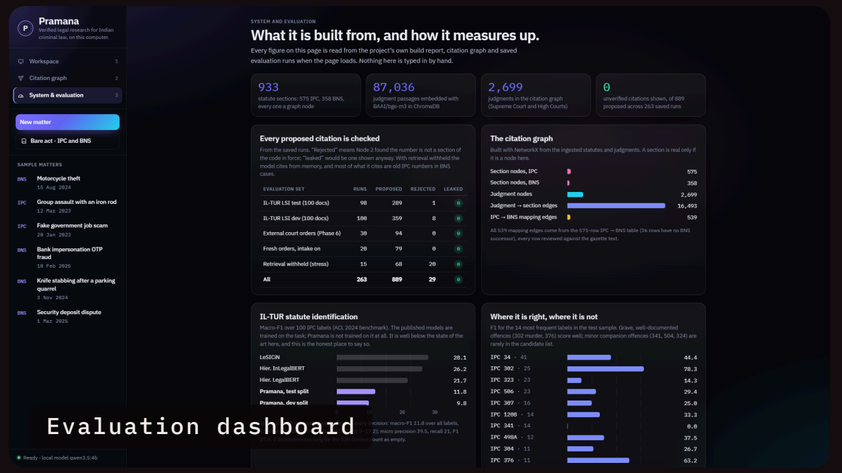
Live pipeline view



Report: supported, doubted, included



Citation-graph view



Evaluation dashboard

The motorcycle case through every stage.

Router	**BNS**	15 Aug 2024 ≥ 1 Jul 2024
Retrieval	**14 candidates**	303(1) first: fused 0.0320, text rank 3, vote rank 2
Node 2	**verified**	303 is a node of the BNS · 0 rejected
Node 3	**3 precedents**	via IPC 379: Telangana 0.686 · HP 0.676 · Delhi 0.658
Node 4	**1 argument**	dishonest intention not proved; quote verified verbatim

Guardrail	**pass**	closest judgment 0.686 ≥ 0.576
Node 1	**proposes BNS 303**	from the candidate list
Node 2b	**supported**	ingredients quoted from the facts, both passes
Node 5	**IPC 379: 158**	76 allowed · 76 dismissed · 1 partly · 5 other · IPC 378: 8 / 3 · BNS 303: suppressed
Timing	**≈ 71 s**	unload 6.4 · guardrail 9.9 · retrieval 0.4 · graph 55.0

One shared state, filled node by node.

`case_facts, offence_date`	input	from the lawyer
`era`	router	"ipc" or "bns"; the only code Node 2 verifies against
`retrieved`	retrieval	ranked candidates Node 1 reads
`candidate_sections`	Node 1	every round's proposals, tagged by round
`verified_sections, unconfirmed, retry_count`	Node 2	verified, rejected with the reason, retries used (≤ 2)
`applicability`	Node 2b	supported or doubted, with quotes and rules
`precedents`	Node 3	graph-linked judgments, IPC bridge for BNS
`analytics`	Node 5	SQL counts, suppressed below 5 judgments
`defenses`	Node 4	counter-arguments, then the final check

code · language model · graph lookup

How each result number was produced.

TEST	SET	HOW IT WAS RUN	RESULT
Citation gate	10 real + 10 invented numbers	verify_section on each, both codes	20 / 20
Routing	10 cases × 3 runs	code chosen and every verified section's code	30 / 30
Stress test	15 dev cases	retrieval withheld; model cites from memory	20 / 68 caught · 6 / 6 recovered
Real court orders	30 fresh Patna HC orders (CC-BY-4.0)	never read before scoring	13 / 30 right section
IL-TUR LSI	100 dev + 100 test documents	decisions on dev; test run once at the end	9.8 / 11.8 macro-F1
Embedder bake-off	15 hand-written cases	each searched only against its own code	0.533 vs 0.20 recall@3
Held-out sets	16 · 10 · 12 cases	written before each change, scored once	14/16 · 6/10 · 8/12

Hand-written sets were authored with AI assistance and are not lawyer-reviewed; the external orders and IL-TUR are independent. Records: docs/evaluation_results.md.